

Coding & Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/jaswanthtangudu111-stack/Rising-Waters
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Bootstrap 5, XGBoost, Scikit-learn
Key Features Implemented	Flood prediction, XGBoost model, Flask web app, Risk analysis, Safety recommendations, Responsive UI
Pending / Incomplete Features	Live weather API integration, Interactive map visualization (Future Scope)
Setup / Run Instructions	Install requirements → Run python app.py → Open http://127.0.0.1:5000

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Partial
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project successfully implements an AI-powered flood prediction system using the XGBoost Machine Learning algorithm. The application follows a modular architecture with a Flask backend and responsive web interface, providing accurate flood probability prediction and risk analysis.